AGRICULTURE OF INDIA

Part I — Irrigation in India

1. Meaning and importance of irrigation

Irrigation is the artificial application of water to agricultural land to support crop growth when rainfall is inadequate, irregular or poorly distributed. It reduces dependence on the monsoon, permits multiple cropping, improves yields and helps cultivate water-intensive crops.

Irrigation is particularly important in arid and semi-arid regions, including Rajasthan, Rayalaseema and large parts of Telangana. However, excessive irrigation without proper drainage can cause waterlogging, soil salinity and groundwater depletion.

2. Major sources of irrigation

A. Wells and tubewells

Groundwater is extracted through open wells and tubewells using electric or diesel pumps. Traditional water-lifting devices include the Persian wheel (Rahat), moat or charsa, and dhekli. Modern borewells are especially important in groundwater-dependent agricultural regions.

Geographical relevance: Punjab, Haryana, western Uttar Pradesh, Gujarat, Andhra Pradesh and Telangana. Over-extraction is a major concern in several semi-arid districts.

B. Canal irrigation

Canals convey surface water from rivers, reservoirs and dams to agricultural fields. They may be perennial or inundation canals. Canal irrigation supports large command areas but depends on water availability, distribution efficiency and maintenance.

Examples: Indira Gandhi Canal in Rajasthan; Bhakra canal system in north-western India; Nagarjuna Sagar and Godavari irrigation systems in the Telugu states.

C. Tank irrigation

Tanks store monsoon runoff in natural or constructed depressions and distribute water to nearby fields. They are particularly suited to hard-rock terrain where groundwater storage is limited and surface runoff can be captured.

State focus: Telangana's traditional chain of tanks, including village irrigation tanks, and tank systems across Rayalaseema and coastal Andhra Pradesh.

D. Other irrigation methods

Lift irrigation, river-lift schemes, check dams, farm ponds and micro-irrigation supplement the three principal sources. Drip irrigation delivers water close to plant roots, while sprinklers distribute water through controlled sprays.

Particularly relevant for horticulture, orchards, vegetables, cotton and water-stressed districts.

3. Updated national data on irrigation

The old percentages in the notes—wells and tubewells 57%, canals 31%, and tanks 6%—should not be retained as current figures.

According to the Government of India's Land Use Statistics at a Glance, 2023–24 (provisional), the distribution of net irrigated area by source was as follows.

Source of irrigation

Tubewells 48%

Canals 21%

Other wells 15%

Tanks 4%

Other sources 12%

4. Irrigation: geographical distribution and problems

Groundwater irrigation: Dominant in many parts of north-western India and increasingly important in peninsular India. It allows flexible irrigation but can lower the water table when extraction exceeds recharge.

Canal irrigation: Important in the Indo-Gangetic plains and major river-

valley projects. Its effectiveness depends on canal maintenance, equitable distribution and the timing of water releases.

Tank irrigation: Historically significant in the Deccan Plateau, where seasonal streams, undulating relief and hard-rock geology favour runoff storage.

Micro-irrigation: Drip and sprinkler systems improve water-use efficiency, although their effectiveness depends on crop choice, maintenance, energy access and farm economics.

Analytical point: Irrigation development should be assessed not merely by the area irrigated, but also by water-use efficiency, groundwater sustainability, crop productivity and the equity of access between farmers.

5. Andhra Pradesh and Telangana: irrigation focus

Andhra Pradesh

The Godavari and Krishna river systems, canal networks, reservoirs and lift-irrigation schemes support agriculture across the state. The Godavari delta and Krishna delta are important rice-growing regions. Rayalaseema faces greater rainfall variability and water stress, making reservoir management, groundwater recharge and efficient irrigation particularly important.

Telangana

The Godavari and Krishna river basins, reservoirs, lift-irrigation projects and traditional tanks shape the state's agricultural geography. The Kaleshwaram Lift Irrigation Project and Mission Kakatiya are important topics for state examinations. Lift irrigation expands access to surface water but involves substantial pumping, energy, maintenance and financial costs.

State-PSC distinction: APPSC questions often benefit from connecting irrigation with the Godavari–Krishna deltas, the Polavaram project, Rayalaseema drought vulnerability and coastal aquaculture. Telangana PSC questions should connect tank irrigation, the Deccan Plateau, the Godavari–Krishna basins, lift irrigation and groundwater management.

Part II — Cropping seasons in India

6. Kharif, Rabi and Zaid crops

Kharif season — monsoon crops

Crops are generally sown with the onset of the south-west monsoon, mainly in June–July, and harvested from September to November, depending on the crop and region. Rice, maize, jowar, bajra, cotton, jute, groundnut and soybean are important examples.

Rabi season — winter crops

Crops are generally sown in October–December and harvested in March–May. Wheat, barley, gram, mustard, lentil and peas are important examples. Many rabi crops benefit from cool growing conditions and relatively dry weather during harvesting

Zaid season — short summer season

Zaid crops are grown between the rabi and kharif seasons, mainly during the hot summer months. They depend substantially on irrigation in many regions. Watermelon, muskmelon, cucumber, vegetables and fodder crops are common examples.

7. Important distinctions for examinations

Cropping seasons are not identical across India. Sowing and harvesting dates vary with latitude, altitude, monsoon onset, irrigation and crop duration.

Rice is mainly a kharif crop, but it is also grown during rabi and summer in irrigated regions, particularly in southern and eastern India.

Wheat is predominantly a rabi crop and is closely associated with winter rainfall, irrigation and the alluvial plains of north India.

Gram and mustard are important rabi crops. Their geographical distribution is influenced by cool weather, soil conditions and moisture availability.

Sugarcane is a long-duration crop whose planting and harvesting schedules vary by region. It should not be forced into a simple kharif–rabi classification.

Some crops, such as maize and pulses, can be grown in more than one season depending on local agro-climatic conditions.

Memory aid for students: Kharif = monsoon; Rabi = winter; Zaid = summer interval.

Part III — Agricultural revolutions in India

8. Green Revolution

The Green Revolution refers to the rapid rise in agricultural production associated with high-yielding varieties, improved irrigation, chemical fertilisers, pesticides, mechanisation and modern agricultural practices. In India, its most prominent early impact was on wheat production during the mid-to-late 1960s, followed by major gains in rice.

Dr. M. S. Swaminathan is widely recognised as the Father of the Green Revolution in India. American agronomist Norman Borlaug played a major role in developing high-yielding wheat varieties internationally. The term “Green Revolution” was popularised by William S. Gaud in 1968.

Geographical impact: Punjab, Haryana and western Uttar Pradesh benefited greatly from wheat-based agricultural intensification. Irrigated areas of other states also benefited, but the gains were uneven.

Benefits: Higher crop yields, improved food security, reduced dependence on food imports and expansion of modern agricultural infrastructure.

Limitations: Regional inequality, groundwater depletion, soil degradation, fertiliser imbalances, pesticide-related problems and the relative neglect of some rain-fed crops.

9. Blue Revolution

The Blue Revolution refers to the expansion and modernisation of fisheries and aquaculture. It includes fish breeding, pond aquaculture, hatcheries, feed management, cold chains, processing, improved fishing infrastructure and scientific management of aquatic resources.

India's fisheries sector is increasingly driven by inland aquaculture, alongside marine fisheries. The term is broader than simply increasing marine catches.

10. Operation Flood — White Revolution

Operation Flood was launched in 1970 as a major dairy-development programme based on producer cooperatives, milk collection, processing, marketing and rural dairy infrastructure. Dr. Verghese Kurien was its principal architect and is widely known as the Father of the White Revolution in India.

The programme helped link village milk producers with urban markets and strengthened cooperative dairy institutions. India remains the world's largest milk producer.

11. Other important agricultural revolutions

Revolution Main sector or product

Yellow Revolution Oilseeds

Golden Revolution Often associated with horticulture and honey; usage varies by source

Silver Revolution Eggs and poultry

Pink Revolution Often associated with meat, poultry processing or onion, depending on context

Black Revolution Petroleum production in traditional exam terminology; sometimes linked with ethanol blending

Grey Revolution Fertilisers

Brown Revolution Usage varies; often linked with leather, non-conventional agriculture or certain plantation products

Exam caution: The colour terminology is not fully standardised across all publications. For a disputed term, follow the definition used in the relevant official syllabus or source rather than treating every label as universally fixed.

Part IV — Updated agricultural production data

12. National foodgrain production

For a current PPT, use the final estimates for 2024–25 as the latest final crop-year dataset identified in the official material reviewed. The Ministry has also published third advance estimates for 2025–26, which

should be used when discussing provisional/current production trends.

INDIA · FINAL ESTIMATES · 2024–25

Total foodgrains 357.73 Mt - Total oilseeds 42.99 Mt Rice- 150.18 Mt

Wheat- 117.95 Mt, Maize- 43.41 Mt, Pulses- 25.68 Mt

Mt = million tonnes. Source: Ministry of Agriculture and Farmers Welfare, final estimates released 20 November 2025.

Other important figures for 2024–25 were:

Nutri/coarse cereals: 63.92 million tonnes.

Millets, reported as Shri Anna: 18.59 million tonnes.

Sugarcane: 454.61 million tonnes.

Cotton: 297.24 lakh bales, each weighing 170 kg.

Jute and mesta: 88.02 lakh bales, each weighing 180 kg.

Soybean: 15.27 million tonnes.

Groundnut: 11.94 million tonnes.

Rapeseed and mustard: 12.67 million tonnes.

Conceptual distinction: Foodgrain production includes cereals and pulses. Oilseeds, sugarcane, cotton, jute and horticultural crops are separate categories. Maize is included in cereals and should not be added again to total foodgrains.

13. Crop-wise geographical distribution

The following table gives the major geographical associations to learn; exact rank and share must be matched to a specified agricultural year.

Crop Important producing regions and Geographical explanation

Rice West Bengal, Uttar Pradesh, Telangana, Andhra Pradesh, Punjab, Odisha, Chhattisgarh Warm conditions, water availability and varied monsoon and irrigation regimes

Wheat Uttar Pradesh, Madhya Pradesh, Punjab, Haryana, Rajasthan Cool growing season, fertile soils and irrigation

Maize Karnataka, Madhya Pradesh, Bihar, Maharashtra, Telangana and Andhra Pradesh Adaptability to varied rainfall

and growing seasons

Jowar Maharashtra, Karnataka, Telangana Drought tolerance and suitability to semi-arid regions

Bajra Rajasthan, Uttar Pradesh, Gujarat, Haryana, Maharashtra Heat tolerance and adaptation to low rainfall

Pulses Madhya Pradesh, Maharashtra, Rajasthan, Uttar Pradesh, Karnataka Semi-arid agriculture, crop rotation and residual soil moisture

Chickpea/gram Madhya Pradesh, Maharashtra, Rajasthan Cool, relatively dry rabi conditions

Groundnut Gujarat, Rajasthan, Andhra Pradesh, Tamil Nadu, KarnatakaWarm climate, suitable soils and both rain-fed and irrigated cultivation

Soybean Madhya Pradesh, Maharashtra, Rajasthan Monsoon rainfall and black-soil regions

Cotton Gujarat, Maharashtra, Telangana, Rajasthan, Punjab, Haryana Warm conditions, suitable soils and a long frost-free growing season

Sugarcane Uttar Pradesh, Maharashtra, Karnataka, Tamil Nadu Long growing period, water availability and established processing industries

Jute West Bengal, Bihar, Assam Humid climate, high rainfall and alluvial soils

Tea Assam, West Bengal, Tamil Nadu, Kerala Humid conditions, rainfall, suitable slopes and plantation infrastructure

Coffee Karnataka, Kerala, Tamil Nadu Tropical highland conditions, shade and well-drained soils

Natural rubber Kerala, Tripura, Karnataka, Tamil Nadu Warm, humid climate and high rainfall

Coconut Kerala, Tamil Nadu, Karnataka, Andhra Pradesh Tropical climate and coastal cultivation

Spices Kerala, Karnataka, Andhra Pradesh, Telangana, Tamil Nadu and other suitable regions Crop-specific climate, soils and

irrigation

14. Rice

Rice is a major staple food and an important irrigated and monsoon crop. It requires warmth, adequate moisture and, for many varieties, substantial water during parts of its growth cycle. It is cultivated in both humid eastern India and irrigated north-western and southern regions.

India's rice geography is diverse: West Bengal and Odisha have humid monsoon conditions; Punjab and Haryana rely heavily on irrigation; the Krishna and Godavari deltas support irrigated paddy; and Telangana has developed a major rice-producing economy.

The official Economic Survey's 2023–24 table placed Telangana first by rice production, followed by Uttar Pradesh and West Bengal. This is a useful historical reference, not a permanent ranking.

15. Wheat

Wheat is primarily a rabi crop. It benefits from cool conditions during vegetative growth and relatively dry, sunny weather as the crop matures. The Indo-Gangetic plains and parts of central India are major wheat-growing regions.

Uttar Pradesh, Madhya Pradesh, Punjab and Haryana are particularly important. Wheat cultivation is less dominant in much of peninsular India because the winter climate and cropping systems differ from those of the northern plains.

16. Millets and coarse cereals

Jowar, bajra, ragi and other millets are important in rain-fed and semi-arid agriculture. They generally have lower water requirements than paddy and can be valuable in regions exposed to rainfall variability.

Rajasthan is particularly associated with bajra; Maharashtra and Karnataka with jowar; and Karnataka with ragi. Telangana's dryland agricultural systems also make millets relevant to climate-resilient agriculture.

Millets are nutritionally valuable, and their importance has grown in discussions of crop diversification, climate adaptation and food security.

17. Pulses

Pulses provide protein and contribute to soil fertility through biological nitrogen fixation. They are often grown in rain-fed areas, in crop rotations or using residual soil moisture.

Madhya Pradesh, Maharashtra, Rajasthan, Uttar Pradesh and Karnataka are important pulse-producing regions. Chickpea is particularly associated with rabi cultivation, while tur (pigeon pea) is predominantly a kharif crop in much of India.

UPSC linkage: Low pulse availability can contribute to food-price pressures because domestic production, yields and imports influence market supply.

18. Oilseeds

Groundnut, soybean, rapeseed-mustard, sunflower, sesame and other oilseeds form a diverse group. Their growing seasons and geographical distribution differ considerably.

Gujarat is strongly associated with groundnut; Madhya Pradesh and Maharashtra with soybean; and Rajasthan with mustard. The National Mission on Edible Oils–Oilseeds and other interventions aim to strengthen domestic production and reduce import dependence.

19. Cotton

Cotton is a tropical and subtropical fibre crop that requires warmth and a sufficiently long frost-free growing season. It is grown under both rain-fed and irrigated conditions.

Gujarat, Maharashtra and Telangana are important cotton-producing states. Black soils are favourable in parts of the Deccan Plateau, although cotton is also cultivated in other soil types.

Environmental issues: Excessive pesticide use, water stress, pest resistance and the economics of input-intensive cultivation are important analytical dimensions.

20. Sugarcane

Sugarcane is a long-duration, water-demanding crop. Uttar Pradesh is a major producer in northern India, while Maharashtra and Karnataka are

important in the peninsular region.

The geographical pattern reflects climate, irrigation, sugar mills, transport, harvesting logistics and the regional economics of sugar production. Sugarcane also links agriculture with ethanol blending, bioenergy, water policy and food-versus-fuel debates.

21. Jute and plantation crops

Jute is a fibre crop concentrated in the humid alluvial plains of eastern India, particularly West Bengal. Tea is associated with Assam, north Bengal and southern hill regions; coffee is concentrated in the southern highlands, especially Karnataka; and natural rubber is strongly associated with Kerala and other humid tropical regions.

These crops illustrate the relationship between agricultural specialisation, physical geography, labour, processing and export markets.

Part V — Oilseeds, horticulture, spices and allied agriculture

22. Horticulture

Horticulture includes fruits, vegetables, flowers, plantation crops, spices and several other high-value crops. It is increasingly important for farm diversification, nutrition, exports and rural employment.

India's horticultural production was estimated at approximately 352.23 million tonnes in the second advance estimates for 2023–24. Fruits were estimated at 112.63 million tonnes and vegetables at 204.96 million tonnes. These are estimates for that year, not current 2025–26 totals.

23. Important fruit crops

Mango: Grown across tropical and subtropical India, including Maharashtra, Uttar Pradesh, Andhra Pradesh and Telangana.

Banana: Important in Maharashtra, Tamil Nadu, Andhra Pradesh, Gujarat and other irrigated or humid regions.

Apple: Strongly associated with Jammu and Kashmir and Himachal

Pradesh, where temperate conditions favour cultivation.

Citrus: Grown in several regions, with production shaped by climate, irrigation and varietal suitability.

Papaya and guava: Widely cultivated in warm regions, including peninsular India.

24. Important spices

Spice Important geographical association

Chilli Andhra Pradesh, Telangana, Karnataka and Maharashtra

Turmeric Telangana, Andhra Pradesh, Maharashtra and Tamil Nadu

Cardamom Kerala, Karnataka and Tamil Nadu

Black pepper Kerala and Karnataka

Ginger Kerala, Meghalaya and other humid regions

Kerala is traditionally called the Spice Garden of India. The term reflects its historic and contemporary association with spice cultivation and trade, rather than implying that every spice is produced predominantly there.

25. Onion and potato

Onion is cultivated in several states, particularly Maharashtra, Madhya Pradesh, Karnataka and Gujarat. Its prices are sensitive to seasonal supply, storage capacity, transport costs and crop losses.

Potato is a major food crop in Uttar Pradesh, West Bengal, Bihar, Gujarat and other states. It is important for food security, processing industries, cold-storage infrastructure and agricultural marketing.

26. Cashew and coconut

Cashew is grown in coastal and tropical regions, including Maharashtra, Andhra Pradesh, Odisha, Kerala and Karnataka. Its production and processing are linked to exports and regional agro-industries.

Coconut is a tropical plantation crop grown widely in Kerala, Tamil Nadu, Karnataka and Andhra Pradesh. It supports food, coir, oil, household industries and value-added products.

Part VI — AP and Telangana: high-priority agricultural geography

27. Andhra Pradesh: agricultural profile

Andhra Pradesh combines coastal delta agriculture, rain-fed cultivation, horticulture, livestock, inland aquaculture and marine fisheries. Its coastline, river deltas, reservoirs and diverse agro-climatic regions make agriculture a major part of its geographical identity.

The Krishna and Godavari deltas are particularly important for irrigated paddy. Coastal districts also support coconut, horticulture, aquaculture and other commercial crops. Rayalaseema has greater exposure to rainfall variability and drought risk.

28. Andhra Pradesh: crops to remember

Crop or sector Important association

Rice Krishna–Godavari deltas and irrigated coastal regions

Chilli Guntur region and other commercial-crop belts

Tobacco Traditional cultivation in selected coastal and inland districts

Groundnut Rayalaseema and other rain-fed areas

Banana and horticulture Irrigated and commercially specialised districts

Aquaculture Coastal districts, brackish-water areas and inland aquaculture clusters

Milk, poultry and eggs Commercial livestock and integrated supply chains

The exact leading district varies by crop and agricultural year. For APPSC, learn the crop–district associations together with river systems, soils, irrigation and agro-climatic conditions.

29. Telangana: agricultural profile

Telangana lies predominantly on the Deccan Plateau. Agriculture is shaped by the south-west monsoon, the Godavari and Krishna river basins, black and red soils, tanks, groundwater and large irrigation projects.

Rice and cotton are major commercial crops. Maize, pulses, soybean, chilli, turmeric, horticulture and other crops contribute to regional agricultural diversity. The expansion of irrigation has altered cropping

patterns in several areas, while groundwater sustainability and crop choice remain important policy issues.

30. Telangana: crops to remember

Crop or sector	Geographical significance
Rice	Irrigated tracts and areas benefiting from surface-water projects
Cotton	Major commercial crop in many semi-arid and black-soil areas
Maize	Grown in several districts under monsoon and irrigated conditions
Turmeric	An important commercial crop, especially associated with the Nizamabad region
Chilli	Commercial cultivation in selected districts
Pulses	Important in rain-fed and mixed-cropping systems
Horticulture	Increasing relevance for diversification and farm incomes
Inland fisheries	Reservoirs, tanks and aquaculture systems

Critical state-level comparison: Andhra Pradesh has a strong coastal-delta and aquaculture dimension. Telangana's agricultural geography is more closely associated with the Deccan Plateau, tank systems, semi-arid conditions and irrigation-led changes in cropping patterns.

Part VII — Livestock, dairy and fisheries

31. Milk production and the White Revolution

India is the world's largest milk producer. According to Basic Animal Husbandry Statistics 2024, national milk production reached 239.30 million tonnes in 2023–24. Uttar Pradesh led with 16.21% of production, followed by Rajasthan at 14.51%, Madhya Pradesh at 8.91%, Gujarat at 7.65% and Maharashtra at 6.71%.

Milk production is influenced by livestock population, breed quality, fodder availability, veterinary services, cooperative institutions, chilling facilities and access to markets.

AP–Telangana focus: Dairy is important to rural livelihoods in both states. For state-specific rankings and output, use the latest state livestock statistics rather than extrapolating from national figures.

32. Fish production and the Blue Revolution

India's total fish production reached approximately 197.75 lakh tonnes in 2024–25, according to provisional national figures. Inland production was 151.60 lakh tonnes and marine production 46.15 lakh tonnes.

Andhra Pradesh is particularly important in aquaculture. In 2024–25, it recorded 55.39 lakh tonnes of total fish production, comprising 48.88 lakh tonnes of inland production and 6.51 lakh tonnes of marine production. Telangana recorded 4.77 lakh tonnes, all classified as inland production in the reported table.

APPSC and Telangana PSC — facts to remember

- Andhra Pradesh has a major role in inland aquaculture, shrimp farming and seafood exports.

- Telangana's fisheries sector depends on inland water resources, including tanks and reservoirs.

- Fish production is not the same as marine fish production: inland aquaculture can account for a large share of national output.

- Fisheries connect agriculture with nutrition, rural employment, exports, cold chains and water quality.

33. Poultry and eggs

India produced an estimated 142.77 billion eggs in 2023–24. Andhra Pradesh contributed 17.85% of national production and Telangana 12.88%, placing both states among India's leading egg-producing states in that dataset.

This is an important state-PSC fact because it demonstrates that the agricultural economy includes livestock and allied activities, not just field crops.

Part VIII — Major agricultural programmes and policy themes

34. Important schemes to cover in the PPT

Programme Main purpose

Pradhan Mantri Krishi Sinchayee Yojana (PMKSY) Irrigation expansion and improved water-use efficiency

Per Drop More Crop Micro-irrigation and water-use efficiency

Pradhan Mantri Fasal Bima Yojana (PMFBY) Crop-risk insurance

PM-KISAN Income support to eligible farmer families

Soil Health Card Scheme Soil testing and nutrient-management advice

National Mission on Natural Farming Promotion of natural farming practices

National Mission on Edible Oils–Oilseeds Strengthening domestic oilseed production

National Food Security Mission Improving production of targeted food crops

Rashtriya Krishi Vikas Yojana Support for state-led agricultural development

Pradhan Mantri Matsya Sampada Yojana (PMMSY) Fisheries and aquaculture development

National Livestock Mission Livestock-sector development and entrepreneurship

Scheme design, eligibility, funding and implementation can change. For exam preparation, update these details from the relevant ministry or state government notifications.

35. Andhra Pradesh: policy topics

Study irrigation projects and command areas, crop diversification, aquaculture regulation, horticulture, agricultural marketing, farmer support programmes and drought management. The state's latest Socio-Economic Survey is a key source for annual agricultural indicators. The AP Finance Department provides official survey editions, including 2024–25.

36. Telangana: policy topics

Prioritise tank restoration, irrigation infrastructure, crop diversification, paddy and cotton cultivation, groundwater management, agricultural marketing, horticulture and livestock. The Telangana government's annual Socio-Economic Outlook provides an official reference for current state-level indicators.

Part IX — UPSC and State PSC analytical notes

37. Why does agricultural productivity vary across India?

Agricultural productivity depends on rainfall, irrigation, soil fertility, temperature, crop varieties, farm size, mechanisation, credit, extension services, storage, transport and market access. High rainfall alone does not guarantee high productivity, and high production does not necessarily imply high yield per hectare.

For examinations, distinguish three concepts:

Area: Land under a particular crop.

Production: Total quantity harvested.

Productivity or yield: Output per unit of cultivated area.

A state may lead in production because it cultivates a larger area, because yields are higher, or because of a combination of both.

38. Why are crop rankings not permanent?

Crop rankings vary with rainfall, drought, floods, pests, irrigation availability, market prices, changes in cultivated area and reporting revisions. Advance estimates may also differ from final estimates.

39. Agricultural challenges

Water stress: Groundwater depletion, inefficient irrigation, water-intensive cropping and competing urban and industrial demands threaten long-term sustainability.

Climate change: Heatwaves, irregular monsoons, droughts, floods and changing pest patterns can affect crop yields and farm incomes.

Soil degradation: Erosion, salinity, nutrient imbalances and declining organic matter can reduce long-term productivity.

Fragmented holdings: Small holdings can constrain mechanisation, investment, storage and bargaining power, although collective farming arrangements and custom-hiring services can help.

Post-harvest losses: Inadequate cold chains, processing facilities, storage and transport reduce the value farmers receive.

Crop concentration: Excessive dependence on a few crops can increase exposure to price shocks, water stress and climate risks.

40. Way forward

- Promote crop diversification based on agro-climatic suitability, water availability and market demand.

- Expand drip and sprinkler irrigation where appropriate, alongside groundwater recharge and watershed development.

- Strengthen soil testing, balanced nutrient management, integrated pest management and climate-resilient seed varieties.

- Improve agricultural storage, cold chains, processing, farmer-producer organisations and market access.

- Encourage millets, pulses, oilseeds and other crops suited to rain-fed agriculture.

- Strengthen crop insurance, agricultural extension, digital advisory services and weather information.

- Integrate livestock, fisheries and horticulture with crop farming to diversify rural incomes.